Linear Algebra

Introduction

Marek Petrik, Adapted from Linear Algebra Lectures by Martin Scharlemann

Mar 14th, 2021

Today: Linear Algebra

Why?

1. **Understand ML Methods**
   - Why different feature sets give the same fit?
   - Why $p - 1$ and not p dummy variables?
   - Why linear regression is the same as orthogonal projection?
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3. **Help Data Processing**

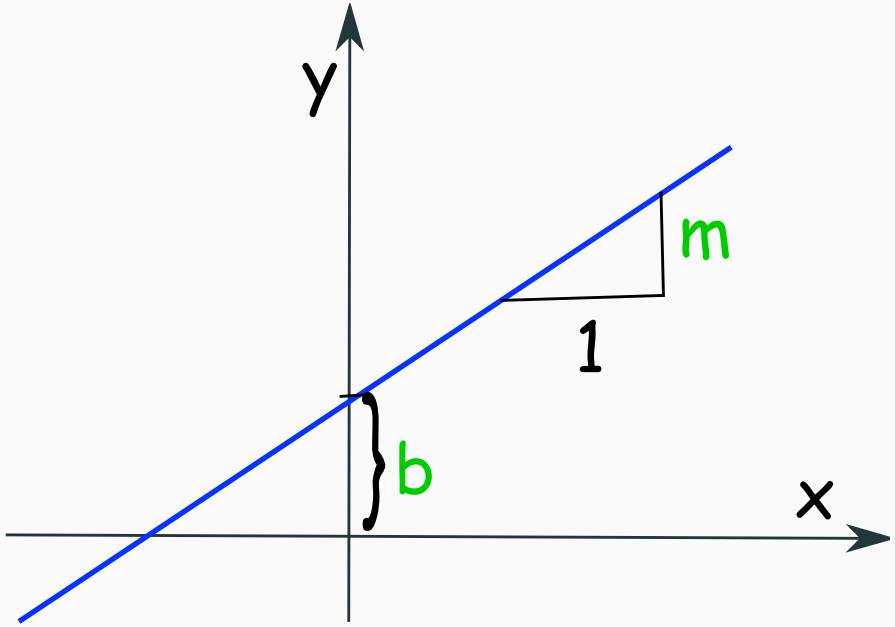
Suggested Linear Algebra Books

1. Strang, G. (2016). Introduction to linear algebra (5th ed.)
   `http://math.mit.edu/gs/linearalgebra/`
   Watch online lectures:
   `https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures/`

2. Hefferon, J. (2017). Linear algebra (3rd ed.).
   Free PDF:
   `http://joshua.smcvt.edu/linearalgebra/`

Linear Equation

Equation of a line:



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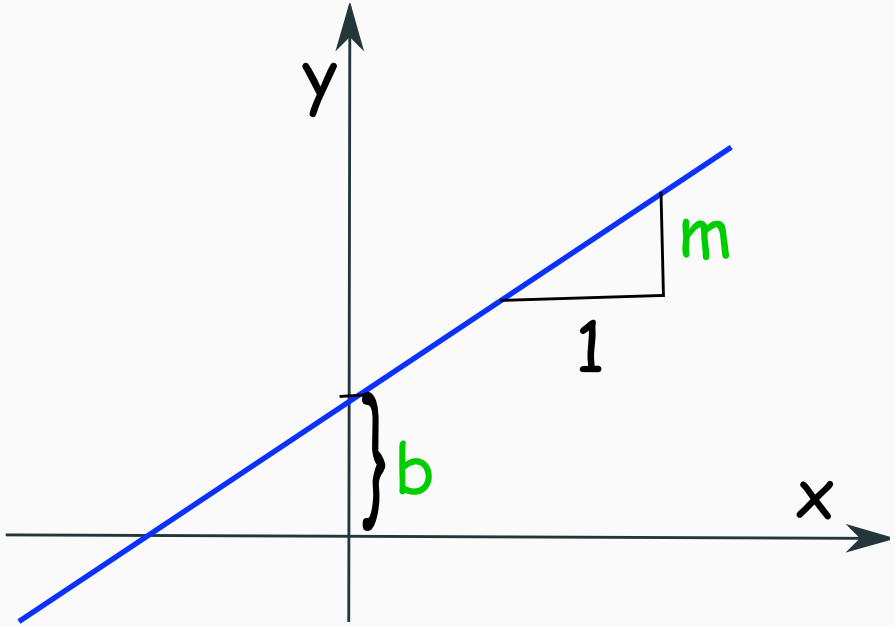

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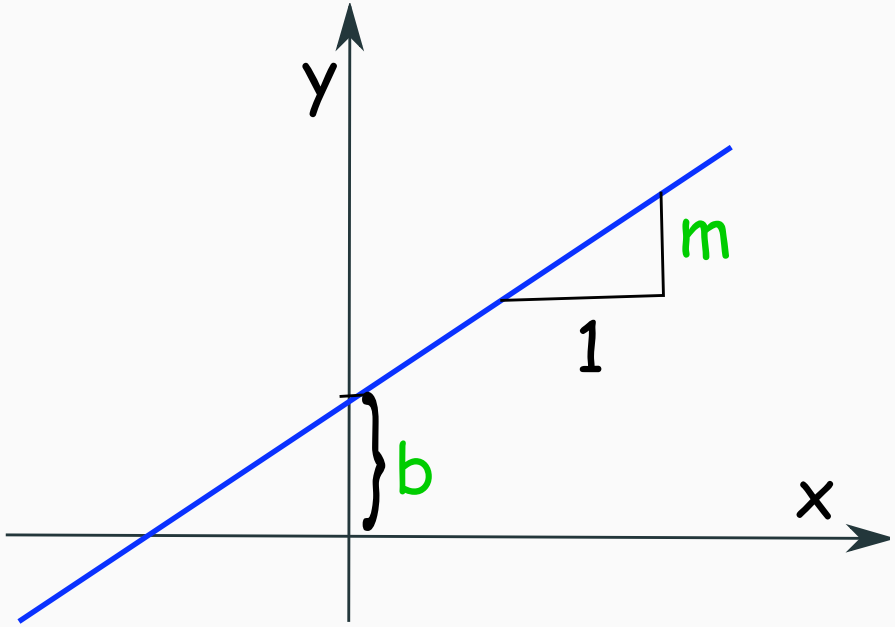

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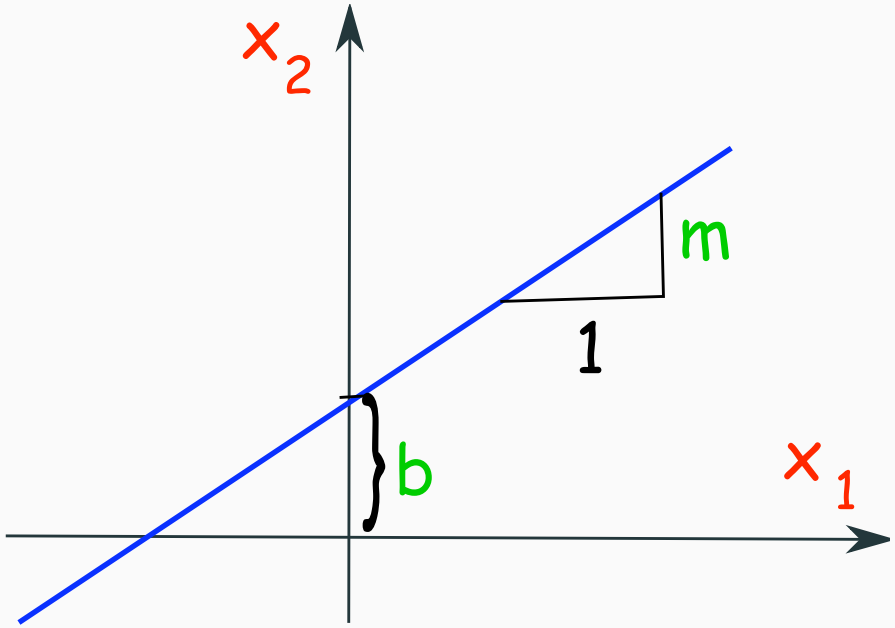

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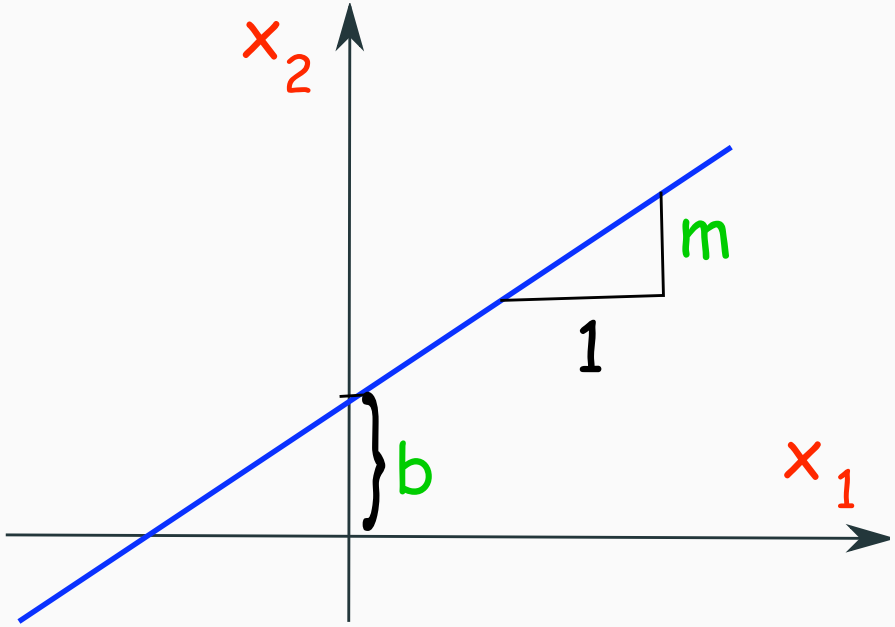

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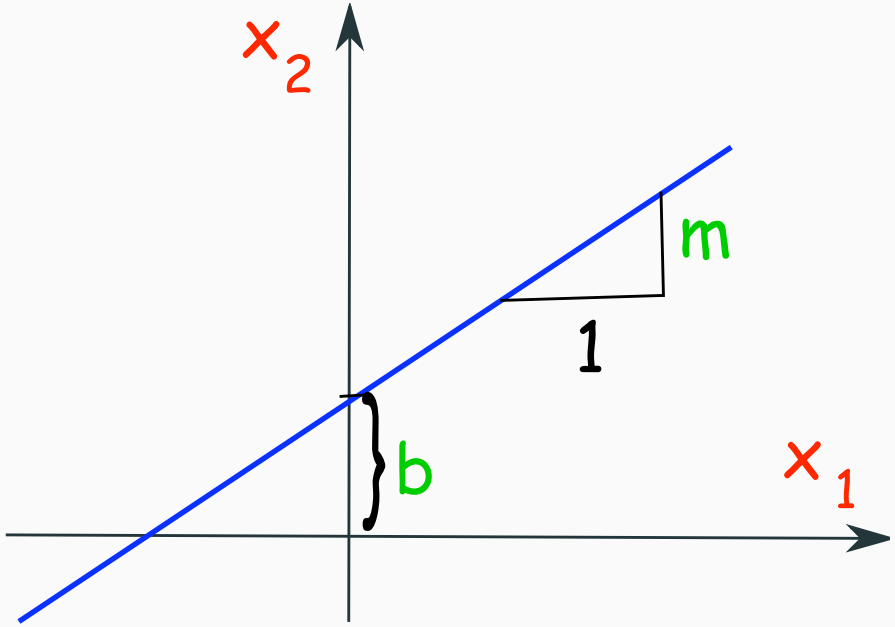

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$$\Downarrow (\Uparrow a_2 \neq 0)$$
$$a_1x_1 + a_2x_2 = b'$$

Linear Equation

Linear equation in variables $x_1, ..., x_n$:

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

where $a_1, ..., a_n$ and b are known in advance.

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Example: For the equation

$$\beta_1 x + (-1)y = -\beta_0$$

of a line, for some β_0, β_1:

The pair x, y is a solution $\iff$ the point (x, y) is on the line.

Systems of linear equations

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That is, all m equations are true when $x_1 = s_1, x_2 = s_2, \ldots, x_n = s_n$.

Example

Conceptual example:

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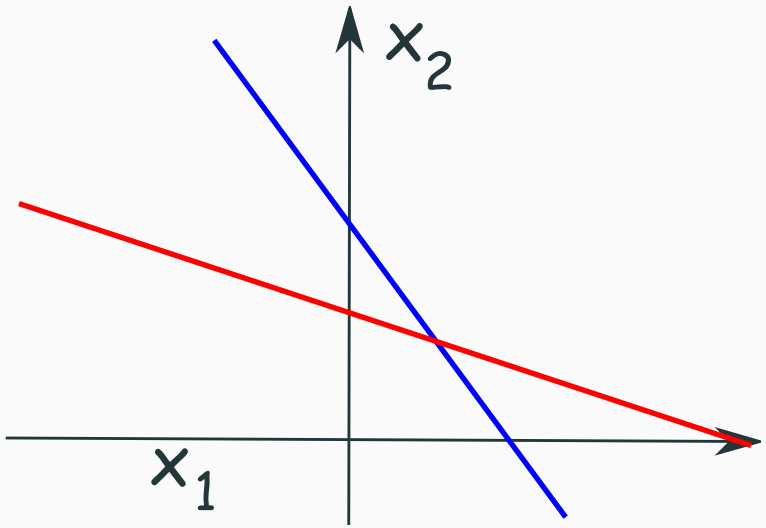

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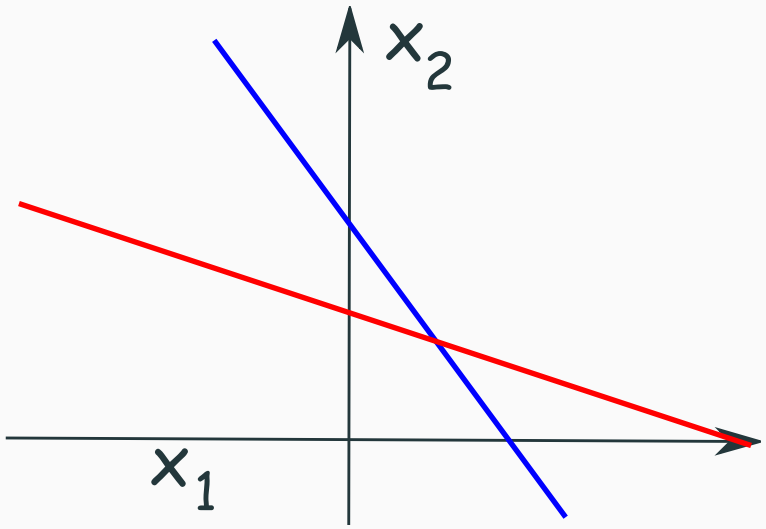
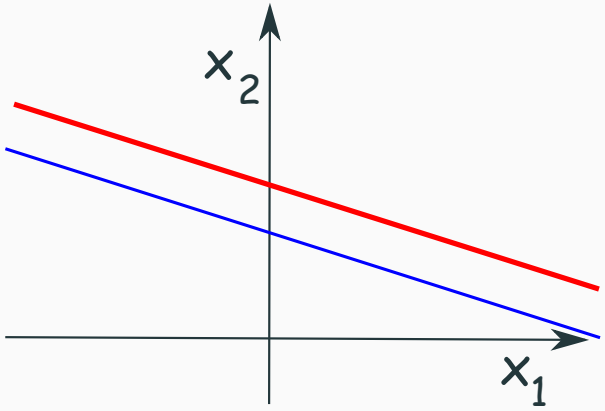


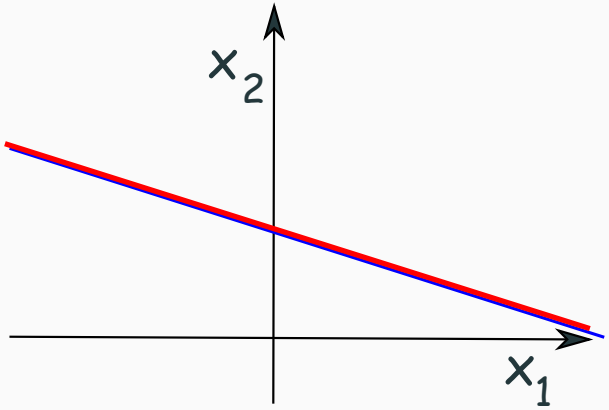
Example:

$$1x_1 + 2x_2 = 3$$
$$2x_1 + 1x_2 = 3$$
$$\Leftrightarrow (x_1, x_2) = (1, 1)$$

Line Configurations

Other possibilities:





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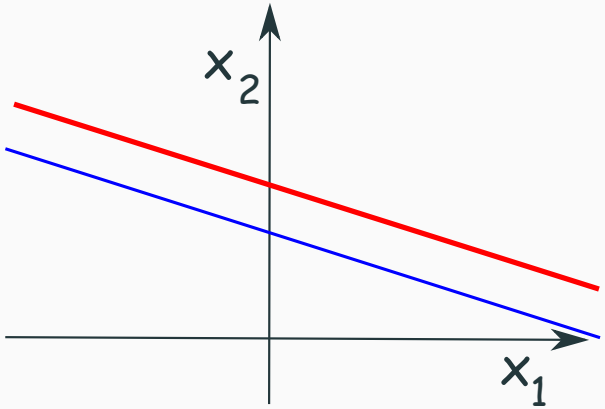

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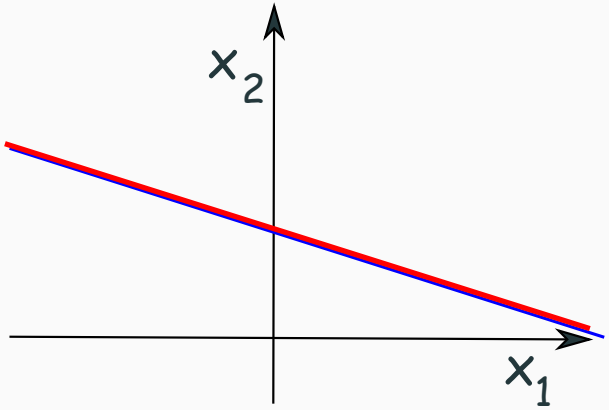
inconsistent



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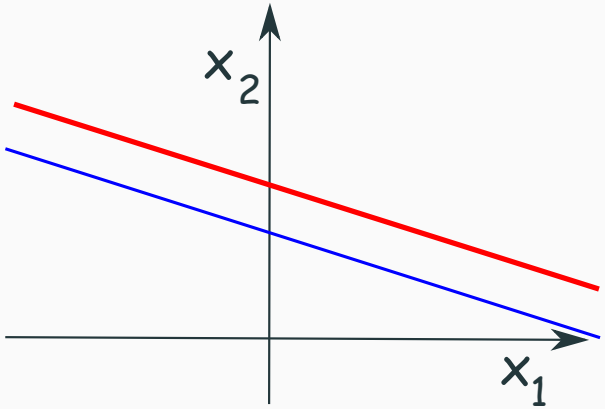

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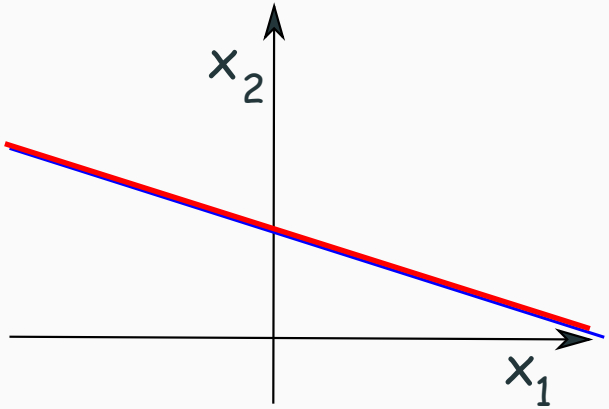

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redundant

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Upshot: There are exactly three possibilities

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- if there are infinitely many solutions, figure out a way to describe them all.

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$m \times (n+1)$ augmented matrix

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Grand strategy: Do this until the equations are easy to solve.

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SOLUTIONS INFINITE

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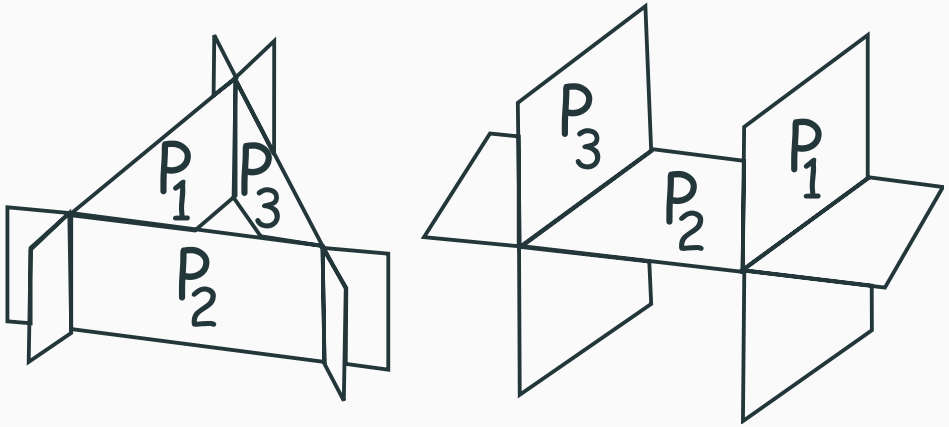

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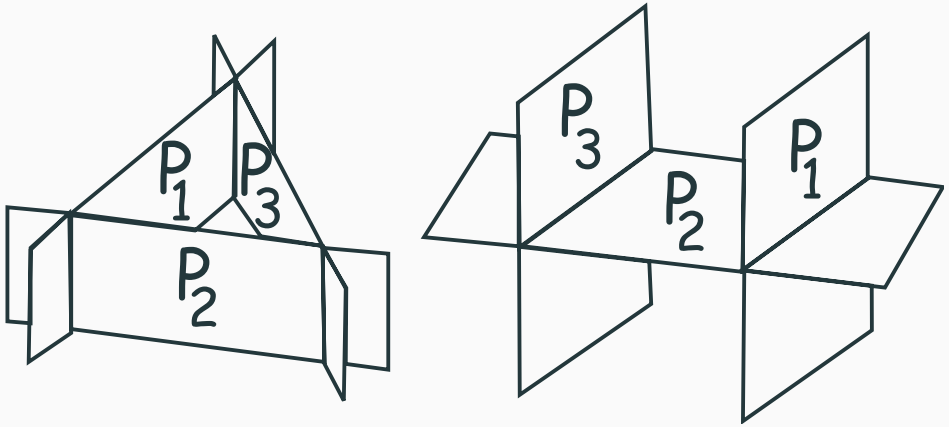

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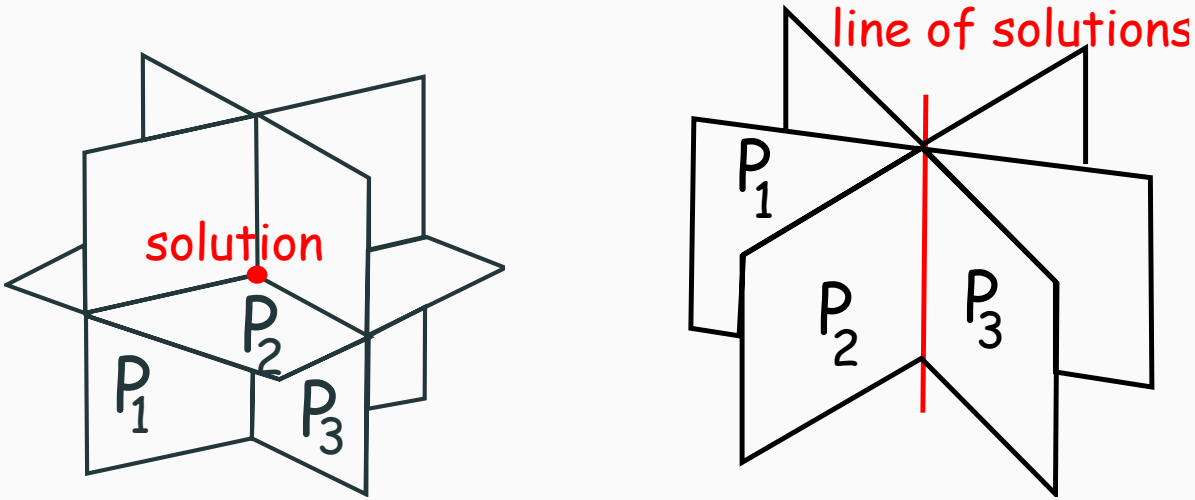

Vectors

Vector

An m-vector [column vector, vector in $\mathbb{R}^m$] is an $m \times 1$ matrix:

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Do not multiply two vectors together like this.

Vector Operations

Examples:

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A) $\begin{bmatrix} 5 \\ -10 \\ -19 \end{bmatrix}$ B) $\begin{bmatrix} 5 \\ -9 \\ 10 \end{bmatrix}$ C) $\begin{bmatrix} 5 \\ 10 \\ 9 \end{bmatrix}$ D) $\begin{bmatrix} 5 \\ -19 \\ -10 \end{bmatrix}$ E) $\begin{bmatrix} 5 \\ 10 \\ 19 \end{bmatrix}$

Picturing vectors

Picturing Vectors

2- and 3-vectors:

- think of vector as arrow from 0
- multiply number with vector via scaling.
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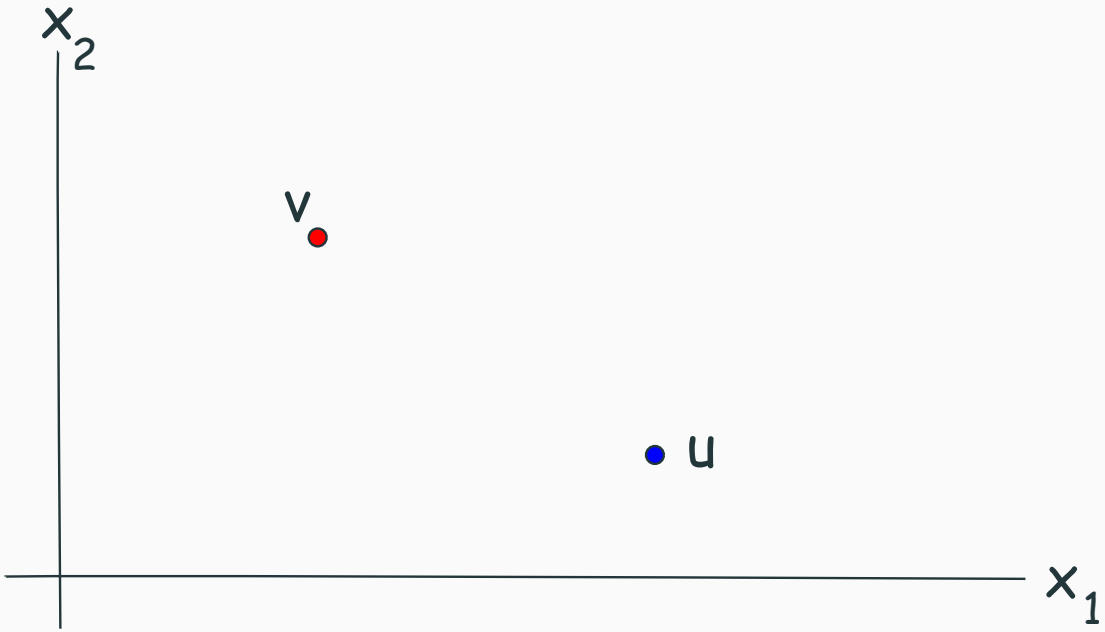

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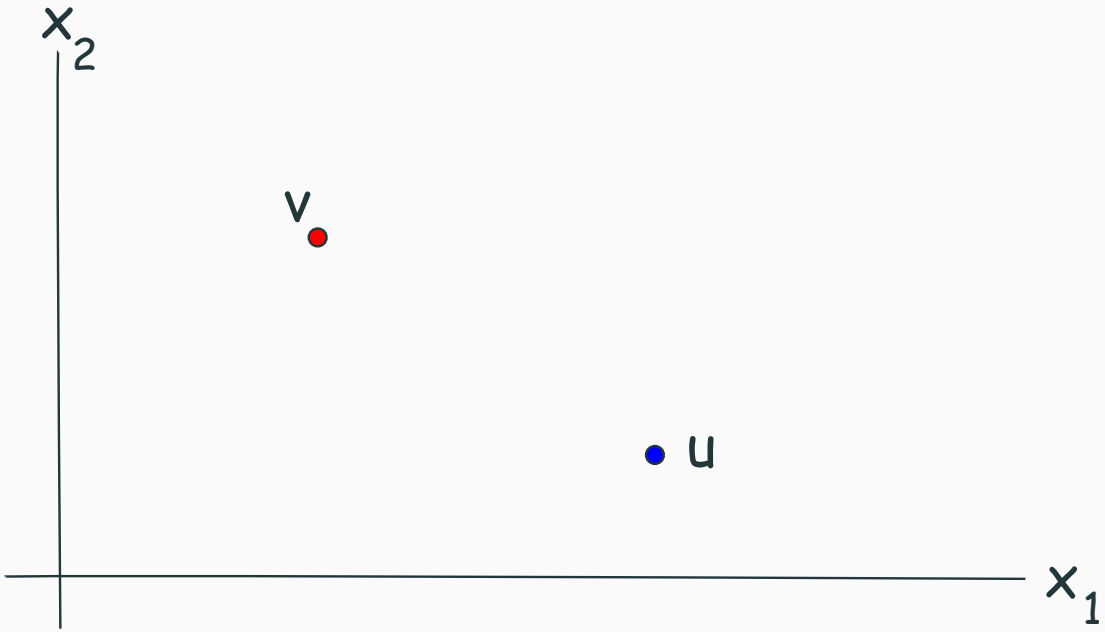

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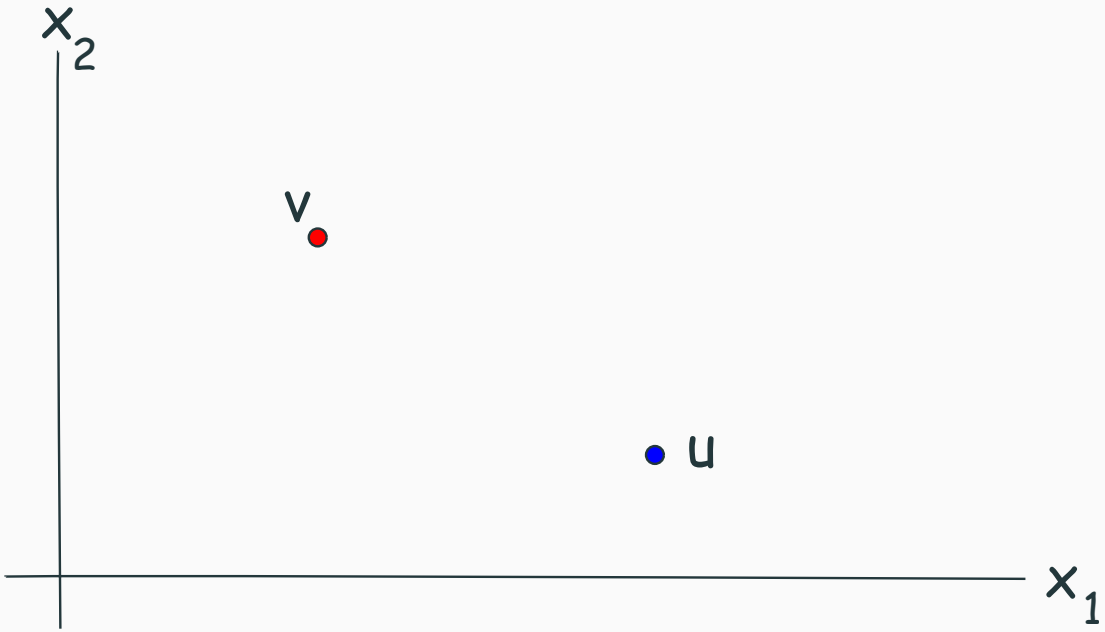

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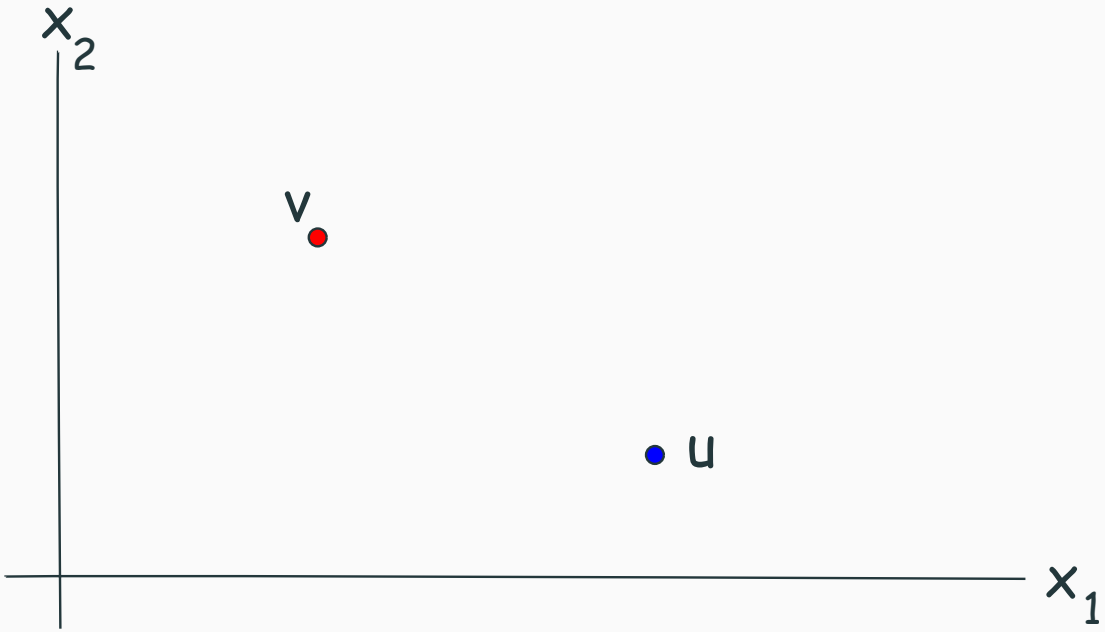

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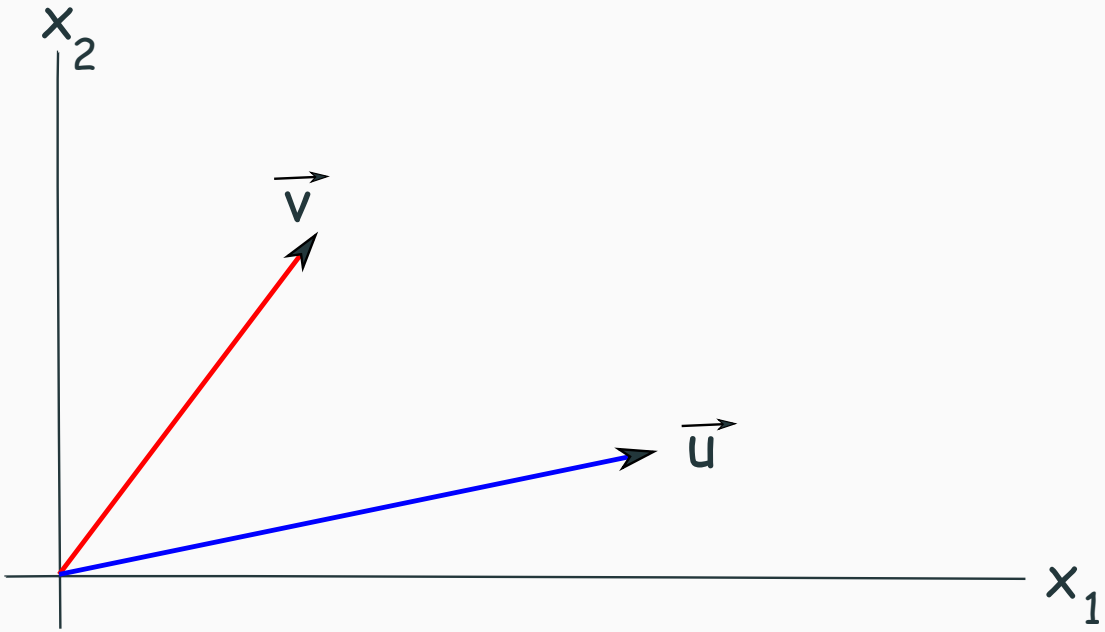

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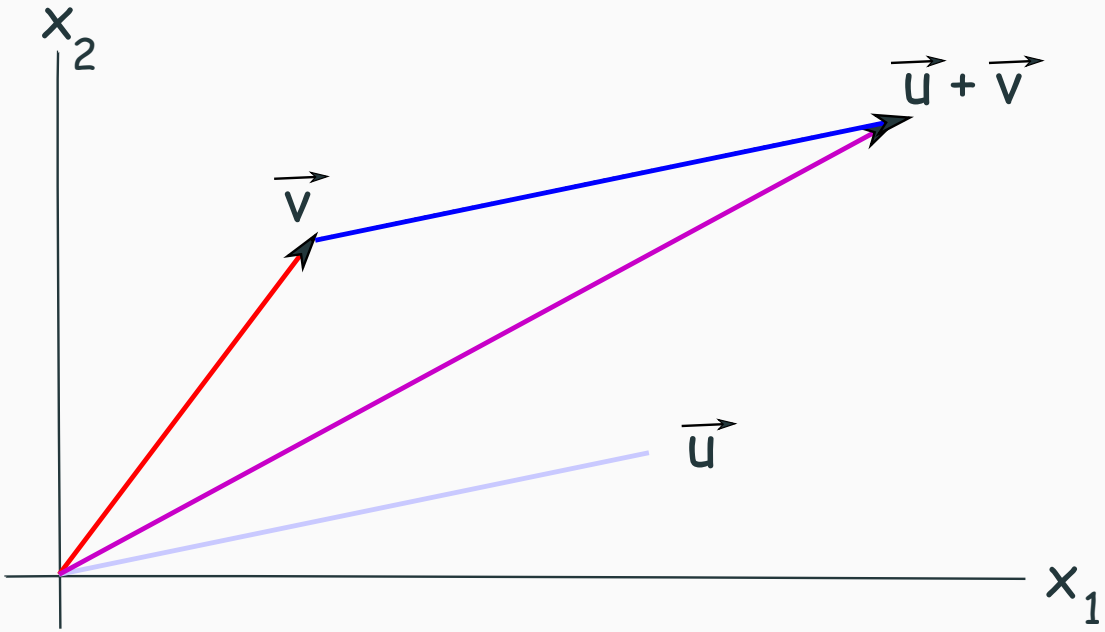

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- think of vector as arrow from 0
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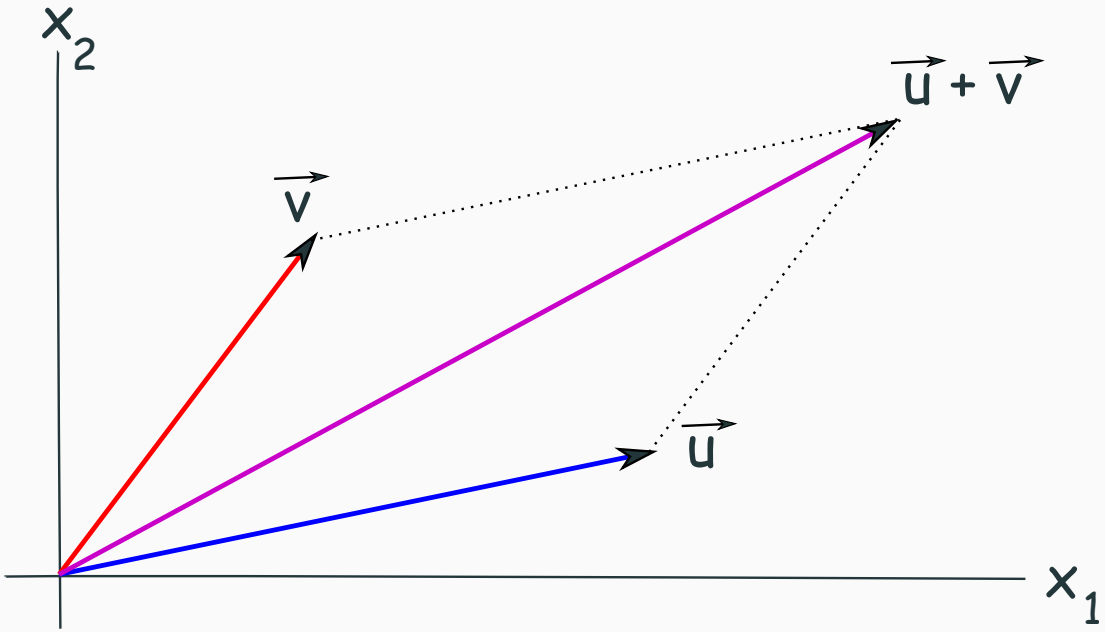

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For a line through a point p, in direction $\vec{d}$, use

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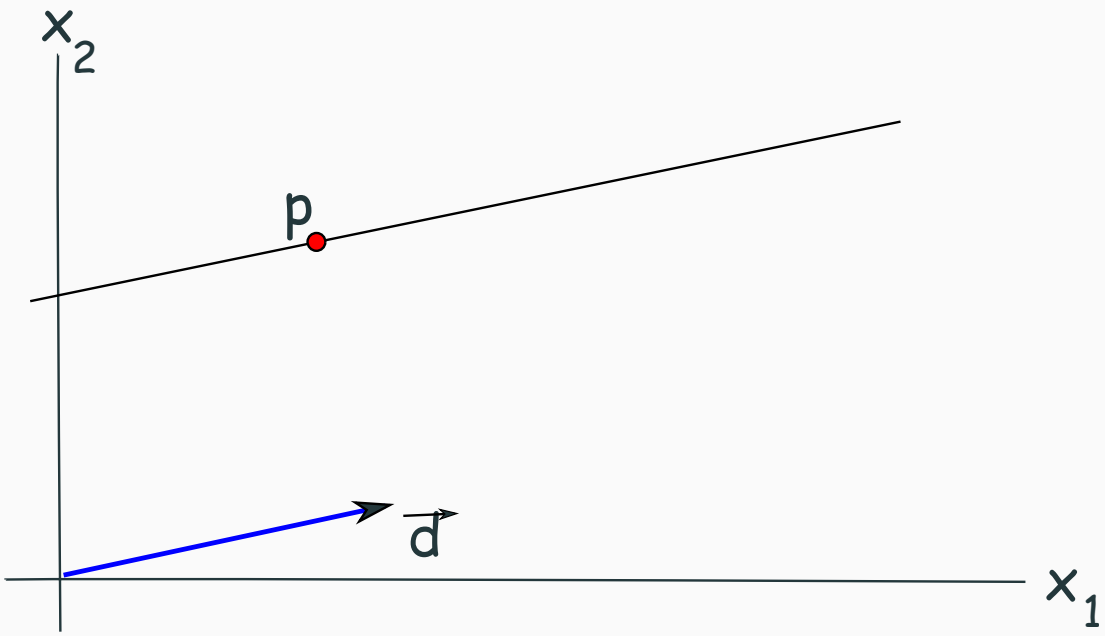

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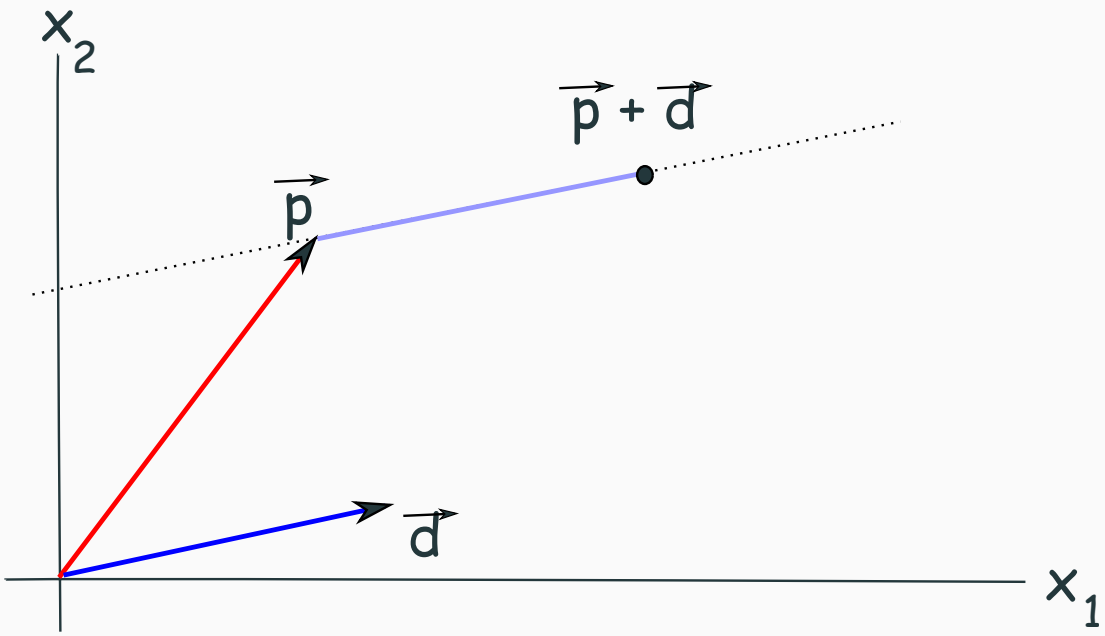

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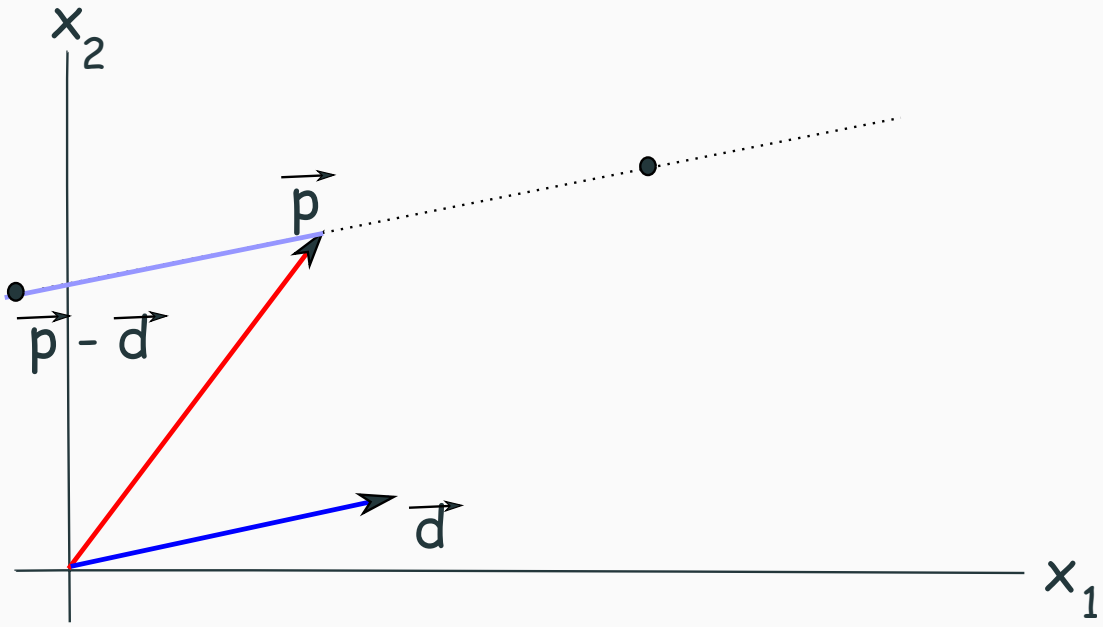

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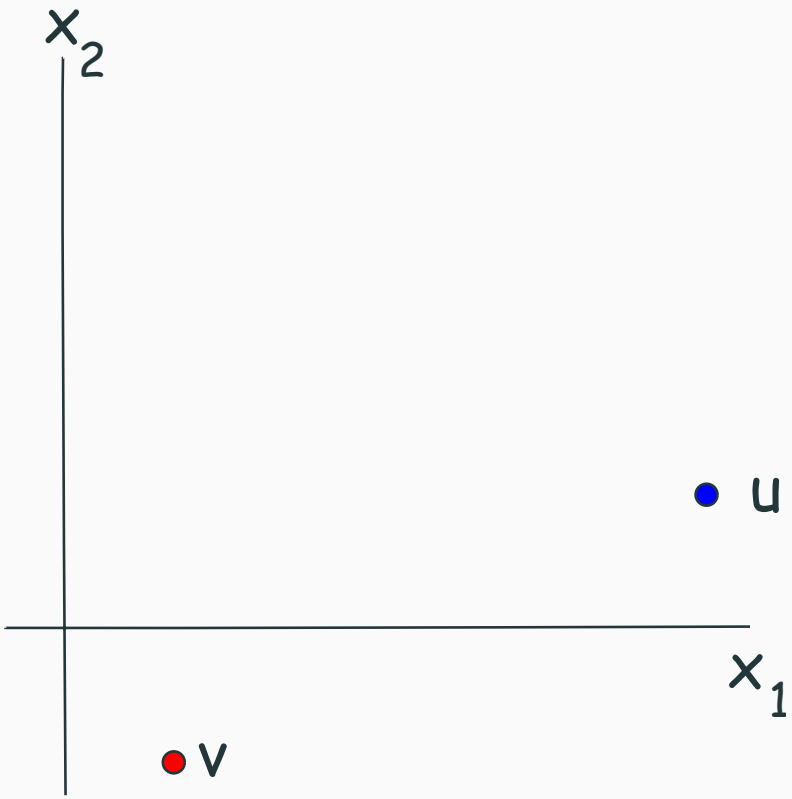

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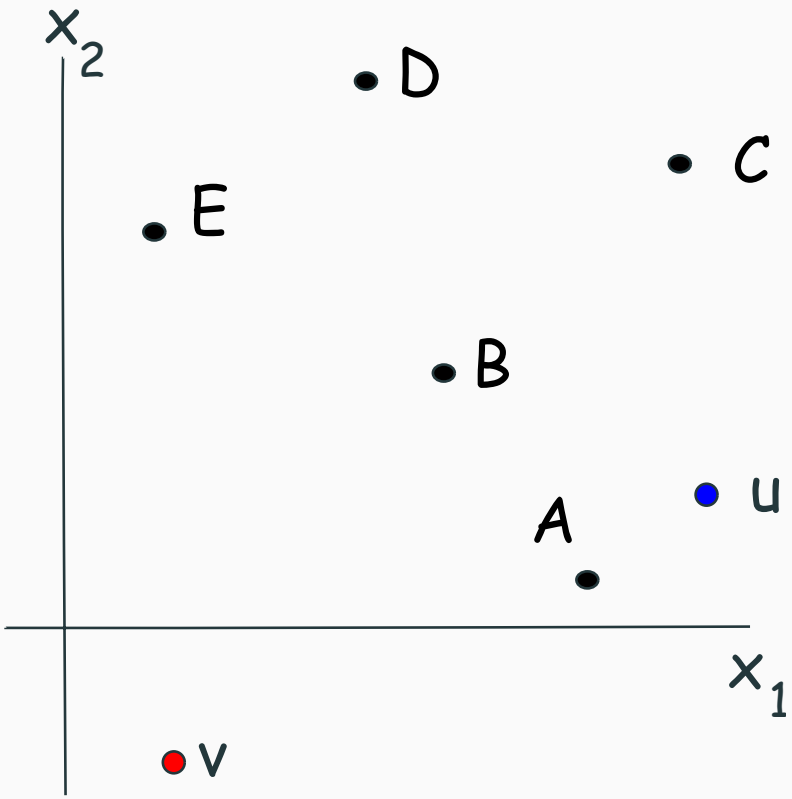

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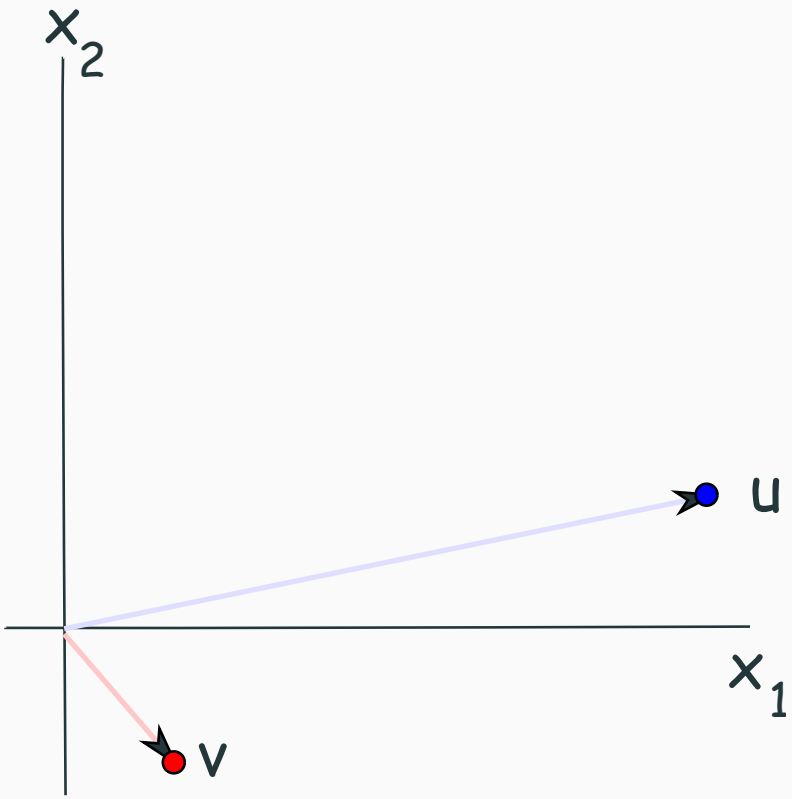

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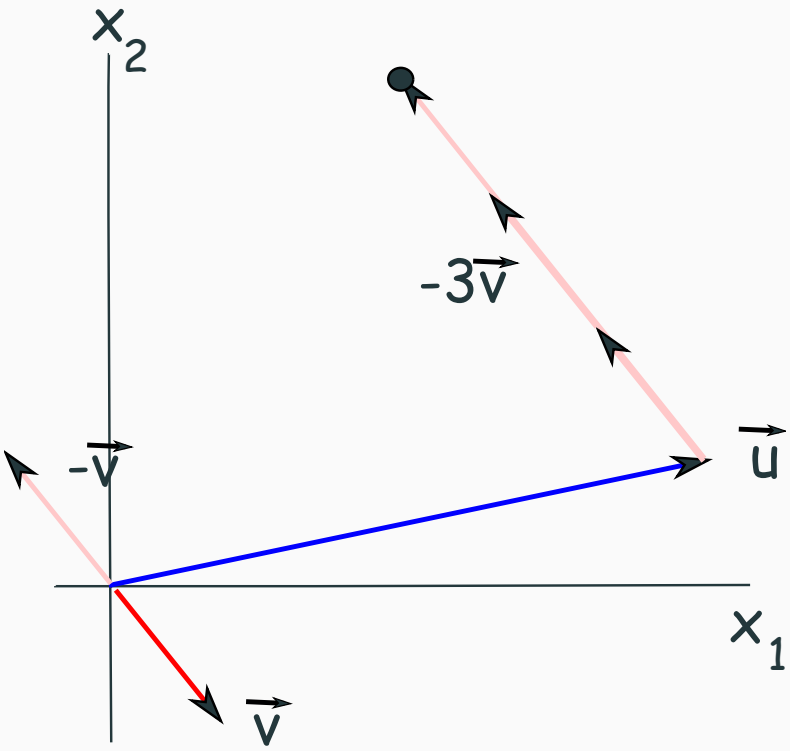

Properties of vectors

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- $1\vec{x} = \vec{x}$

Linear combination

Linear Combination

Definition
Suppose $\{t_1, t_2 \dots t_k\}$ are all real numbers.

The vector

$$\vec{y} = t_1\vec{v}_1 + \cdots + t_k\vec{v}_k$$

is called a linear combination of the vectors $\{\vec{v}_1, \vec{v}_2 \dots \vec{v}_k\}$.

Linear Combination

How to think about solving for $\{t_1, t_2 \dots t_n\}$ in the equation

$$t_1\vec{a}_1 + \cdots + t_n\vec{a}_n = \vec{b}:$$

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Let

$$\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_m \end{bmatrix}; \qquad \vec{a}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \\ a_{mj} \end{bmatrix} \; for\ 1 \leq j \leq n$$

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Then for any $1 \leq i \leq m$, the ith row of the equation becomes:

$$t_1 a_{i1} + t_2 a_{i2} + \cdots + t_n a_{in} = b_i \ or$$

$$a_{i1}t_1 + a_{i2}t_2 + \cdots + a_{in}t_n = b_i$$

Linear Combination

In other words, solving

$$t_1\vec{a}_1 + \cdots + t_n\vec{a}_n = \vec{b}:$$

is the same as solving this system of m linear equations:

$$\begin{aligned} a_{11}t_1 + \ldots + a_{1n}t_n &= b_1 \\ a_{21}t_1 + \ldots + a_{2n}t_n &= b_2 \\ &\vdots \\ a_{m1}t_1 + \ldots + a_{mn}t_n &= b_m \end{aligned}$$

We just learned how to do this!

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Since each $\vec{a}_j$ is a column of i numbers, can just write

$$\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n & \vec{b} \end{bmatrix}$$

Example

Suppose

$$\vec{a}_1 = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 8 \end{bmatrix} \quad \vec{a}_2 = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 8 \end{bmatrix} \quad \vec{a}_3 = \begin{bmatrix} 6 \\ -1 \\ 1 \\ -1 \end{bmatrix} \quad \vec{a}_4 = \begin{bmatrix} 0 \\ 6 \\ 10 \\ 26 \end{bmatrix}$$

and want to find c_1, c_2, c_3, c_4 so that

$$c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 + c_4\vec{a}_4 = \begin{bmatrix} 12 \\ 4 \\ 13 \\ 23 \end{bmatrix}$$

Example

This translates to the system of linear equations whose augmented matrix is

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and so has general solution

$$c_2 = anything,\ c_1 = \frac{3}{2} - c_2, c_3 = 2, c_4 = \frac{1}{2}$$

Compact notation

Matrix Multiplication

Definition
The linear combination

$$x_1\vec{a}_1 + \cdots + x_k\vec{a}_k$$

is abbreviated

$$\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \dots & \vec{a}_k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_k \end{bmatrix}$$

.

Here $\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \dots & \vec{a}_k \end{bmatrix}$ is the matrix A with i^{th} column $\vec{a}_i$.

Matrix Multiplication

Simplest example: each $\vec{a_i} \in \mathbb{R}^1$, i. e. each column in the matrix is just a number:

$$\begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_k \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_k \end{bmatrix} = a_1 b_1 + a_2 b_2 + \dots a_k b_k.$$

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Matrix-vector multiplication

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Think of doing the simple case on each row of the matrix A:

Matrix-vector Multiplication

Apply simple case to first row:

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Further abbreviation:

Use vector notation:

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Matrix-vector Multiplication

Summary: For A an $m \times n$ matrix, and a vector $\vec{x} \in \mathbb{R}^n$, multiplication $A\vec{x}$ is defined and gives a vector in $\mathbb{R}^m$.

Multiplication has two important properties:

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Sample problem from before:

Given vectors $\{\vec{a}_1, \vec{a}_2 \ldots \vec{a}_n\}$ and $\vec{b}$ in $\mathbb{R}^m$, find real numbers $\{x_1, x_2 \ldots x_n\}$ so that

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where $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \dots & \vec{a}_n \end{bmatrix}$. Note: the Matrix-vector multiplication $A\vec{x}$ makes sense only if the number of columns in A matches the number of entries in x.

Linear Independence

How do we put these ideas into math lingo, so we can be precise?

Definition
A set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\}$ in $\mathbb{R}^m$ is linearly independent if and only if the only solution to the equation

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is the solution $x_i = 0$ for $1 \leq i \leq k$.

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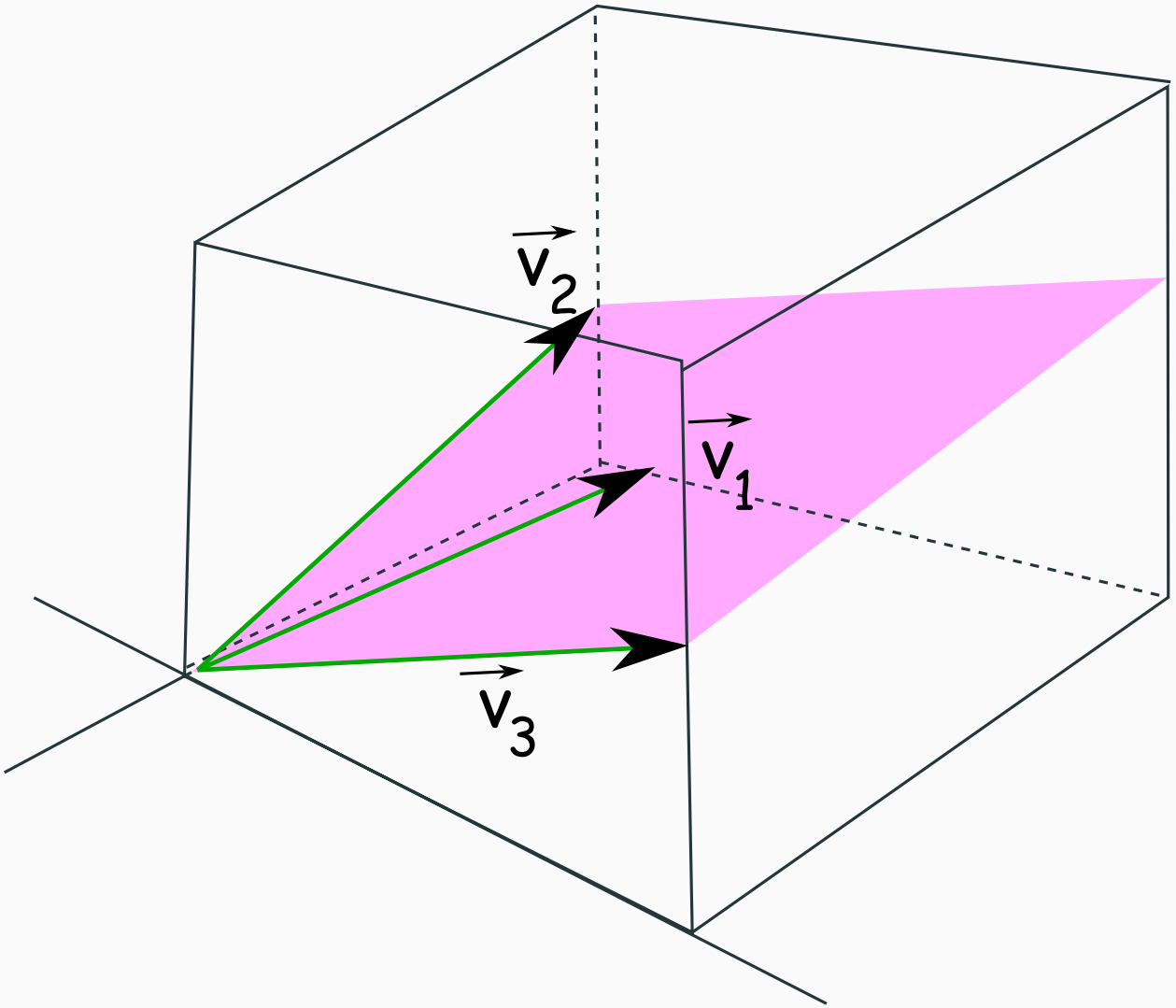
If the set is linearly dependent then span is "thinner" than it has to be; you could even throw some away and not change the span.

In pictures:



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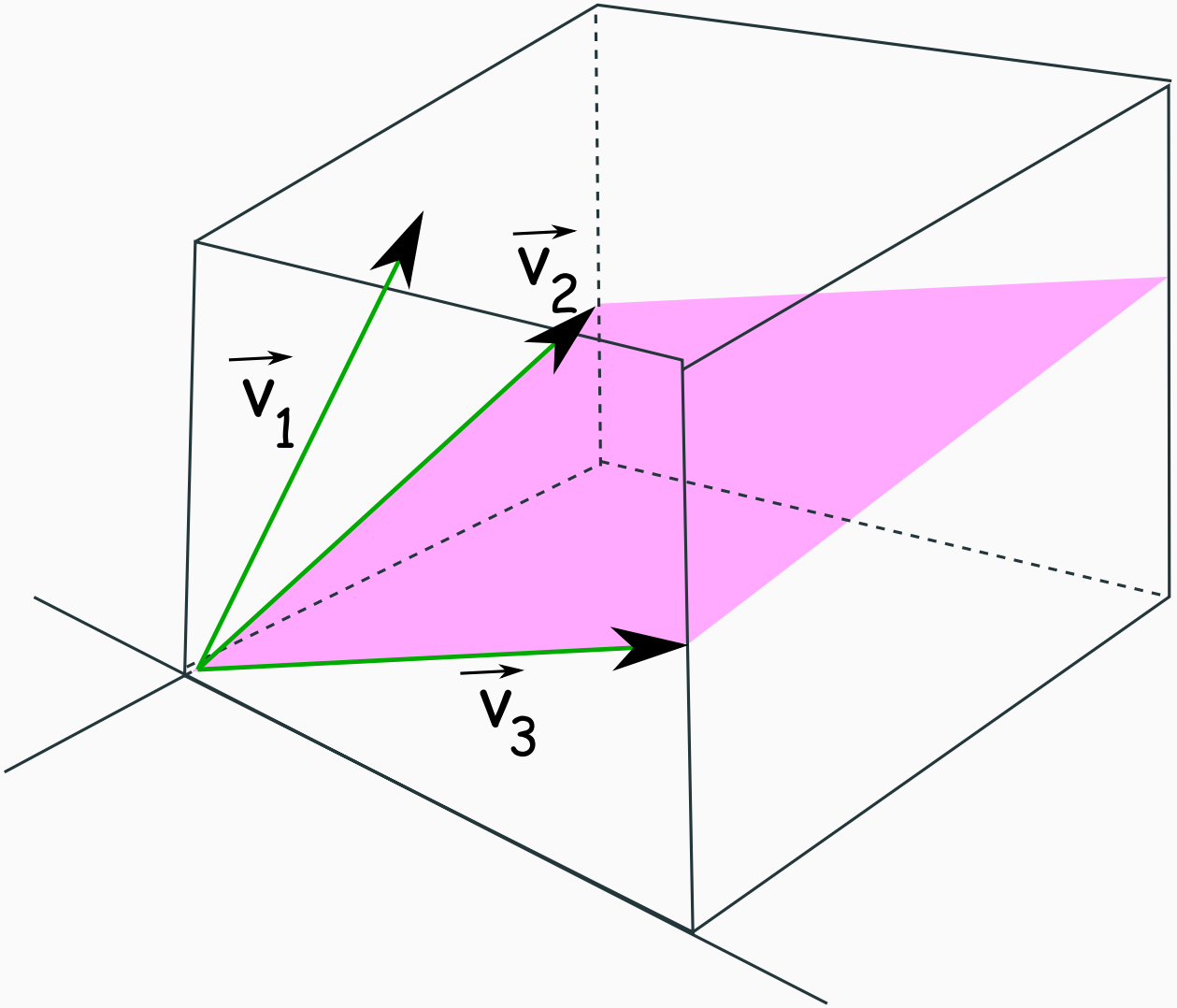

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But then,

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Equivalently: if $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is linearly independent then so is every subset of vectors from $\{\vec{v}_1, \ldots, \vec{v}_k\}$.

How to determine (in)dependence

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There are no free variables if and only if linearly independent